

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most current WHO malaria guideline identified as of 29 April 2026:

WHO guidelines for malaria, v9.0 — 13 August 2025. The NCBI Bookshelf record lists this as the latest update and states that this version replaces earlier versions, including the 30 November 2024 version; WHO and MAGICapp pages direct users to the online platform for the most current recommendation text and revision dates ([NCBI Bookshelf](#), [WHO Global Malaria Programme](#), [MAGICapp](#)).

Important currency note: A WHO Malaria Policy Advisory Group document indicated that updated recommendations on malaria treatment and chemoprevention were planned for **July 2026**, but as of **29 April 2026** that update had not yet occurred ([WHO MPAG document](#)).

Exactly 5 WHO malaria-related guideline documents synthesized

No.	WHO guideline / technical guidance document	Publication / update date available	Scope
1	WHO guidelines for malaria, v9.0	13 August 2025	Consolidated WHO recommendations on prevention, vector control, chemoprevention, vaccines, diagnosis, treatment and case management (NCBI Bookshelf PDF).
2	WHO Guidelines for malaria	31 March 2022	Evidence-based recommendations on malaria prevention, vector control, chemoprevention, vaccine, diagnosis and treatment across transmission settings (MAGICapp PDF).
3	WHO Guidelines for malaria	13 July 2021	Earlier consolidated recommendations, including chemoprevention and case management, such as IPTi and SMC in specified African settings (MAGICapp PDF).
4	WHO Malaria Technical Updates	Exact publication date not visible in source; includes 2023 implementation references	Operational guidance across ITNs, IRS, chemoprevention, vaccines, diagnosis, ACTs, severe malaria treatment, community case management and surveillance (WHO Malaria Technical Updates).
5	WHO Malaria Guidelines:	Exact publication date not visible in source; includes	Focused guidance on IPTp, perennial malaria chemoprevention, SMC, IPT in

No.	WHO guideline / technical guidance document	Publication / update date available	Scope
	Preventive Chemotherapies	2023 implementation references	school-aged children, PDMC and MDA (WHO Preventive Chemotherapies PDF).

Required title check: “Meaningful youth engagement...”

A WHO publication with the title “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” exists, but it concerns **abortion care / sexual and reproductive health**, not malaria. It is therefore **not counted** among the five malaria-related WHO guidance documents above ([WHO Western Pacific publication page](#)).

Target population

WHO malaria recommendations apply to:

- People living in malaria-endemic areas.
- Children, pregnant women and other high-risk groups.
- Travellers, migrants and displaced populations exposed to malaria.
- Patients with suspected or confirmed uncomplicated or severe malaria.
- National malaria programmes adapting WHO guidance to local epidemiology, transmission intensity, health-system capacity and antimalarial resistance patterns.

Scope and objectives

The WHO malaria guideline platform aims to provide a single, updated source for recommendations on:

- **Diagnosis:** rapid parasitological confirmation by microscopy or rapid diagnostic tests.
- **Treatment:** uncomplicated malaria, severe malaria, species-specific treatment, ACTs, pregnancy, infants and antimalarial resistance.
- **Prevention:** vector control, chemoprevention, malaria vaccines and surveillance.
- **Programme management:** national adaptation, subnational stratification and implementation according to local transmission and resistance patterns. WHO’s Global Technical Strategy emphasizes that national treatment guidelines should account for local antimalarial drug-resistance patterns and health-system context ([WHO Global Technical Strategy](#)).

1.2 Key Recommendations

GRADE note: Where explicit WHO GRADE labels were available in the accessible source summaries, they are reported directly. Where exact MAGICapp PICO-level labels were not exposed in the accessible excerpts, the table provides a guideline-level GRADE judgement based on the cited evidence type, consistency and clinical certainty; official implementation should verify the relevant MAGICapp evidence profile.

Recommendation	GRADE Evidence Quality	Strength
In all patients with suspected malaria, perform prompt parasitological diagnosis using microscopy or malaria rapid diagnostic test; do not delay testing when malaria is clinically possible.	High	Strong
Results of parasitological diagnosis should be available rapidly; WHO technical updates state results should be available within less than two hours of presentation, and if diagnosis is absent or delayed, suspected severe malaria and high-risk groups should be treated clinically.	Moderate	Strong
Treat uncomplicated malaria with an appropriate artemisinin-based combination therapy (ACT) according to species, pregnancy status, drug resistance, location of acquisition and national policy.	High	Strong
Treat uncomplicated malaria urgently, ideally within 24–48 hours after symptom onset, with treatment choice guided by parasite species, disease severity, pregnancy, comorbidities, drug interactions and resistance patterns.	Moderate	Strong
Hospitalize and urgently treat severe malaria ; parenteral artesunate remains the core severe-malaria treatment approach, with specialist consultation when delayed parasite clearance or artemisinin resistance is suspected.	High	Strong
For severe <i>P. falciparum</i> malaria with suspected artemisinin resistance, consult a malaria specialist; a 2025 treatment recommendations document reports WHO recommendation of artesunate plus quinine combination therapy in that situation.	Low	Conditional
For <i>P. vivax</i> or <i>P. ovale</i> malaria, assess G6PD status before anti-relapse therapy; WHO's 30 November 2024 update added recommendations on near-patient qualitative and semiquantitative G6PD testing and updated primaquine/tafenoquine recommendations.	Moderate	Strong
Provide malaria vaccines in a four-dose schedule for children starting from 5 months of age to reduce malaria disease and burden; countries may give the first dose earlier for operational reasons, with a minimum interval of 4 weeks and a fourth dose 6–18 months after dose 3.	Moderate	Strong
Use malaria chemoprevention strategies according to epidemiologic setting and target group, including IPTp, SMC, PMC, IPT in school-aged children, PDMC and MDA where appropriate.	Moderate	Strong
For seasonal malaria chemoprevention , use monthly amodiaquine plus sulfadoxine–pyrimethamine in areas of highly seasonal transmission where the regimen remains effective, adapted to current WHO age and transmission guidance.	High	Strong

Recommendation	GRADE Evidence Quality	Strength
For post-discharge malaria chemoprevention , consider drug duration of protection, transmission season length and feasibility of successive treatments.	Moderate	Conditional
Deploy indoor residual spraying for malaria prevention and control in children and adults living in areas with ongoing malaria transmission; the 2025 update incorporated evidence for chlorfenapyr and isocycloseram and emphasized reduced DDT use.	Very Low	Strong
Spatial emanators / spatial repellents can be deployed for malaria prevention and control in children and adults in areas with ongoing transmission.	Moderate	Conditional
Maintain antimalarial resistance surveillance using therapeutic efficacy monitoring and, where feasible, molecular markers such as <i>Pfkelch13</i> variants; adapt national treatment policy to local resistance patterns.	Moderate	Strong
National malaria programmes should use local data for subnational stratification and adapt WHO recommendations to local transmission, resistance, vector ecology and health-system capacity.	Moderate	Strong

Key supporting sources: WHO 2025 guideline record and PDF ([NCBI Bookshelf](#), [NCBI Bookshelf PDF](#)); WHO technical and preventive chemotherapy guidance ([Technical Updates](#), [Preventive Chemotherapies](#)); CDC diagnostic and case-management summaries ([CDC Yellow Book](#), [CDC Case Management](#)); Guideline Central summary of 2025 WHO IRS and spatial emanator GRADE labels ([Guideline Central](#)).

1.3 Guideline Development Methodology

Evidence review process

WHO malaria recommendations are maintained as a **living consolidated guideline** in MAGICapp. Updated recommendations display their most recent revision date, and each update is accompanied by a downloadable PDF version on the WHO website or linked repositories ([NCBI Bookshelf](#), [MAGICapp](#)).

The 13 August 2025 update:

- Revised information on **indoor residual spraying**, incorporating evidence for **chlorfenapyr** and **isocycloseram**.
- Maintained the overall IRS recommendation while emphasizing reduced use of **DDT**.
- Added a new recommendation on **spatial emanators / spatial repellents** ([NCBI Bookshelf](#)).

The 30 November 2024 version included:

- Updated malaria vaccine recommendation.
- New near-patient qualitative and semiquantitative G6PD testing recommendations for *P. vivax* and *P. ovale* anti-relapse therapy.

- Updated primaquine recommendations.
- Recommendation on tafenoquine ([ReliefWeb / WHO guideline notice](#)).

Consensus and advisory process

WHO malaria recommendations are informed by systematic evidence review, GRADE-style certainty assessment, expert review and WHO advisory processes, including the Malaria Policy Advisory Group. MPAG documents describe ongoing review of treatment, chemoprevention and vector-control topics and note planned future updates ([WHO MPAG document](#)).

Conflict of interest management

WHO guideline development processes normally require declaration and management of conflicts of interest among guideline panel members, methodologists, evidence reviewers and external contributors. The accessible malaria guideline summaries confirm the WHO/MAGIcapp process but do not expose individual conflict-of-interest forms in the cited excerpts; implementation teams should consult the full MAGIcapp guideline package for recommendation-specific evidence-to-decision documentation.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Time window

- **Latest WHO malaria guideline release used as baseline: 13 August 2025.**
- **Post-guideline evidence window: 14 August 2025 to 29 April 2026.**

Search strategy

Evidence was prioritized from:

- WHO official guideline pages, MAGIcapp, WHO IRIS and WHO-linked technical documents.
- Authoritative public-health institutions, especially CDC.
- Systematic reviews, meta-analyses, evidence reports and clinical-trial methodology publications.
- Recent malaria treatment, chemoprevention, vaccine, vector-control and resistance surveillance literature.

Databases and sources represented

- WHO Global Malaria Programme and MAGIcapp.
- NCBI Bookshelf WHO guideline record.
- CDC malaria clinical and Yellow Book guidance.
- WHO-linked vaccine evidence report.
- Peer-reviewed and academic repositories including Frontiers, AJTMH, PMC and Oxford Centre for Tropical Medicine and Global Health.

Inclusion criteria

Included evidence if it was:

- 1. Published or updated after **13 August 2025**, or clearly relevant to interpreting the latest WHO update.
- 2. Related to malaria diagnosis, treatment, severe disease, chemoprevention, vaccines, vector control or antimalarial resistance.
- 3. Systematic review, meta-analysis, randomized-trial evidence synthesis, formal evidence report, clinical-trial protocol, authoritative guideline or technical recommendation.

Exclusion criteria

Excluded or downgraded if:

- Publication date could not be verified within the post-guideline window.
- Findings were older than the latest WHO guideline and already likely incorporated.
- Evidence was not malaria-specific.
- Evidence did not address diagnosis, treatment, prevention or management.

2.2 Key New Studies and Evidence

Study / evidence source	Design	Key findings	GRADE Quality	Strength / impact
WHO-linked malaria vaccine evidence report for SAGE, 2025	Evidence report using GRADE; pooled randomized-trial evidence	Report states GRADE was used to inform malaria vaccine recommendations; malaria vaccines had broader rollout since 2024, with 22 countries offering malaria vaccines in childhood immunization programmes. Pooled RCT evidence included five randomized controlled trials with follow-up from 15 weeks to 2 years.	Moderate	Supports current WHO vaccine recommendation; no change required
Swiss malaria treatment recommendations, 2025	Expert clinical recommendations aligned with WHO and recent resistance literature	Addresses suspected artemisinin partial resistance and delayed parasite clearance; recommends specialist	Low	Reinforces need for resistance-aware specialist management

Study / evidence source	Design	Key findings	GRADE Quality	Strength / impact
		consultation for severe <i>P. falciparum</i> malaria with suspected resistance and reports WHO recommendation of artesunate plus quinine combination therapy in that situation.		
Frontiers review, 2025, on malaria resistance / PDMC considerations	Narrative / review evidence	States resistance threatens control and elimination; managing resistance requires genotypic and phenotypic monitoring. Reports WHO recommendation to adopt therapeutic efficacy protocols in endemic regions. Also summarizes PDMC considerations including drug duration, transmission season and feasibility of successive treatments.	Low	Supports surveillance and context-specific chemoprevention implementation
Severe malaria core outcome set protocol, 13 April 2026	Protocol for core outcome set development using PRISMA-informed review, qualitative interviews, Delphi survey and consensus meeting	Aims to standardize outcomes for severe malaria treatment trials, reduce reporting bias and improve comparability. Plans two-round Delphi survey with 60–80 participants and consensus meeting.	Very Low for clinical effect; Moderate for trial-methodology relevance	No immediate treatment change; improves future severe-malaria evidence quality

Study / evidence source	Design	Key findings	GRADE Quality	Strength / impact
CDC Yellow Book and CDC malaria case-management guidance	Authoritative public-health clinical guidance	Emphasizes considering malaria in febrile travellers from endemic areas; immediate testing with microscopy and concurrent RDT if faster; severe malaria requires hospitalization and urgent treatment.	Moderate	Reinforces WHO-compatible diagnostic urgency
Guideline Central summary of WHO 2025 update	Guideline summary	Reports WHO 2025 IRS recommendation as strong / very low certainty and spatial emanator recommendation as conditional / moderate certainty .	Directly reported	Supports vector-control implementation according to WHO GRADE labels

Sources: [Malaria vaccines evidence report, 2025](#); [Swiss malaria treatment recommendations 2025](#); [Frontiers review, 2025](#); [Centre for Tropical Medicine and Global Health, 13 April 2026](#); [CDC Yellow Book](#); [Guideline Central](#).

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority
Artemisinin partial resistance in Africa	Emerging <i>Pfkelch13</i> mutations and delayed parasite clearance could undermine ACT effectiveness; evidence is evolving and geographically heterogeneous.	High
Optimal response to suspected ACT failure	Specialist recommendations exist, but comparative clinical trial evidence for prolonged ACT, alternative ACTs or non-artemisinin rescue regimens remains limited.	High
Severe malaria outcomes	The 2026 core outcome set protocol highlights heterogeneity in severe-malaria trial outcomes, limiting comparability and meta-analysis.	High
Chemoprevention drug choice and resistance impact	WHO notes ACTs may be effective for PMC, but evidence is limited on safety, adherence, cost-effectiveness and resistance consequences.	High

Evidence gap	Why it matters	Priority
Post-discharge malaria chemoprevention implementation	RCT/meta-analysis evidence supports benefit, but operational questions remain around regimen, timing, adherence and health-system delivery.	Moderate–High
G6PD testing implementation for radical cure	Near-patient G6PD tests are critical for safer primaquine/tafenoquine use, but implementation quality, access and pharmacovigilance need strengthening.	High
Malaria vaccine schedule optimization	Four-dose schedules are recommended, but real-world effectiveness, coadministration, seasonal timing and combination with SMC require ongoing evaluation.	Moderate–High
Vector-control tools in insecticide resistance settings	IRS insecticide choice, spatial emanators, LLIN durability and vector shifts such as <i>Anopheles stephensi</i> require continued surveillance.	High
Humanitarian emergencies and fragile settings	Malaria control in displaced populations requires operational evidence on diagnosis, treatment supply chains, chemoprevention and vector control.	High

2.4 Updated Recommendations Based on New Evidence

No post-13 August 2025 evidence identified by 29 April 2026 requires overturning the latest WHO malaria guideline. The main implication is **reinforced implementation**, especially for rapid diagnosis, ACT stewardship, resistance surveillance, vaccine rollout, chemoprevention delivery and vector-control adaptation.

Original WHO recommendation / position	New evidence impact through 29 April 2026	Updated recommendation for practice as of 29 April 2026
Use prompt parasitological diagnosis for suspected malaria.	CDC guidance reinforces urgent testing with microscopy and concurrent RDT when faster.	Maintain. Test immediately; do not send initial tests to distant reference labs if this delays treatment decisions.
Treat uncomplicated malaria with ACTs according to national policy and resistance patterns.	No new post-guideline RCT evidence found that changes ACT-first strategy; resistance concerns continue.	Maintain. Use locally recommended ACT; monitor for delayed parasite clearance and treatment failure.
Treat severe malaria urgently and hospitalize.	2026 severe-malaria core outcome set protocol highlights need for standardized future trial outcomes but does not change treatment.	Maintain. Use urgent parenteral therapy and supportive care; adopt standardized outcomes in future trials.

Original WHO recommendation / position	New evidence impact through 29 April 2026	Updated recommendation for practice as of 29 April 2026
Manage suspected artemisinin resistance with specialist input.	2025 treatment recommendations reinforce specialist consultation and note WHO-reported artesunate plus quinine combination therapy for severe disease with suspected resistance.	Strengthen implementation. Consult malaria experts urgently when delayed clearance or resistance is suspected.
Use G6PD testing to guide radical cure for <i>P. vivax</i> / <i>P. ovale</i> .	No later evidence found that changes 2024 WHO update; implementation remains key.	Maintain. Expand near-patient G6PD testing and pharmacovigilance for primaquine/tafenoquine.
Provide malaria vaccines in a four-dose schedule from 5 months of age.	2025 vaccine evidence report supports ongoing rollout and reports 22 countries using malaria vaccines in childhood programmes.	Maintain and scale. Integrate vaccine delivery with EPI and local malaria seasonality where relevant.
Use SMC, PMC, IPTp, PDMC, IPT in school-aged children and MDA according to epidemiology.	Reviews reinforce benefit but highlight implementation, resistance and regimen questions.	Maintain with local adaptation. Use chemoprevention where WHO criteria are met; monitor efficacy and resistance.
Deploy IRS in ongoing transmission settings.	2025 WHO update incorporated chlorfenapyr and isocycloseram evidence; GRADE summary: strong recommendation, very low certainty.	Maintain. Use IRS where programmatically appropriate, with insecticide-resistance management and reduced DDT reliance.
Consider spatial emanators / repellents in ongoing transmission settings.	2025 WHO update added this recommendation; GRADE summary: conditional, moderate certainty.	Adopt selectively. Use in settings where operational feasibility, acceptability and cost-effectiveness are favourable.
Adapt WHO guidance to local resistance and health-system context.	WHO Global Technical Strategy and recent resistance reviews reinforce local surveillance.	Strengthen. Base national policy on therapeutic efficacy studies, molecular surveillance and subnational stratification.

Practical Clinical Algorithm

A. Suspected malaria

1. Assess urgency

- Fever or compatible illness plus exposure to malaria-endemic area.

- Look for severe features: impaired consciousness, seizures, respiratory distress, shock, severe anaemia, jaundice, renal impairment, acidosis, hypoglycaemia, high parasitaemia.

2. Test immediately

- Microscopy where available.
- RDT concurrently if faster.
- Do not delay treatment in severe malaria while awaiting delayed confirmation.

3. Classify

- Uncomplicated malaria.
- Severe malaria.
- Species: *P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, *P. knowlesi*, mixed infection.
- Pregnancy, age/weight, comorbidities and drug contraindications.

B. Uncomplicated malaria

- Treat with nationally recommended ACT.
- Ensure full-course adherence.
- Counsel on warning signs and follow-up.
- If *P. vivax* or *P. ovale*, evaluate for radical cure with primaquine or tafenoquine only after appropriate G6PD assessment.

C. Severe malaria

- Hospitalize immediately.
- Start parenteral antimalarial treatment urgently.
- Manage complications: hypoglycaemia, seizures, anaemia, acidosis, renal failure, respiratory distress and shock.
- If delayed parasite clearance or suspected artemisinin resistance: consult a malaria specialist and notify public-health authorities.

D. Prevention and programme management

- Combine interventions: ITNs/LLINs, IRS where appropriate, chemoprevention, vaccines, prompt diagnosis and effective treatment.
- Use malaria vaccines in eligible children according to WHO schedule.
- Apply chemoprevention strategies by local transmission pattern and target population.
- Maintain resistance surveillance and update national policy accordingly.

Overall Conclusion

As of **29 April 2026**, the latest WHO malaria guideline remains **WHO guidelines for malaria, v9.0 — 13 August 2025**. The guideline supports an integrated malaria strategy: **rapid parasitological diagnosis, ACT-based treatment of uncomplicated malaria, urgent hospital treatment for severe malaria, G6PD-guided radical cure for relapsing malaria, chemoprevention, vaccination, vector control and resistance surveillance**.

The evidence published or identified after the latest WHO release does **not** justify changing WHO's core recommendations before the planned July 2026 treatment and chemoprevention update. It does, however, reinforce several implementation priorities: rapid diagnosis, resistance-aware treatment, expanded vaccine rollout, careful chemoprevention deployment, improved severe-malaria trial outcomes and locally adapted vector-control strategies.